

Signal Conditioning — Bill of Materials

5-Channel Chua Output Conditioning · X7R Memcapacitor Chaos Generator

Schematic: signal_conditioning.kicad_sch Revision: Rev 3 Date: 2026-08-08 Total components: 83

C0G/NP0 mandatory for all signal-path capacitors (C_FB, C_SH). X7R is reserved for power bypass only. Do not substitute.

RESISTORS — 25 TOTAL

Reference	Value	Description	Footprint	Qty	Notes
R1 / R12 / R23 / R34 / R45	1 kΩ	Input series R_IN (CH0-4)	0402	5	1% — Bourns CR0402-FX-1001ELF
R4 / R15 / R26 / R37 / R48	30 kΩ	Divider top R_DIV_H (CH0-4)	0402	5	1%
R5 / R16 / R27 / R38 / R49	10 kΩ	Divider bottom R_DIV_L (CH0-4)	0402	5	1%
R6 / R17 / R28 / R39 / R50	10 kΩ	Sallen-Key R_SK1 (CH0-4)	0402	5	1%
R8 / R19 / R30 / R41 / R52	10 kΩ	Sallen-Key R_SK2 (CH0-4)	0402	5	1%
Resistor subtotal				25	

CAPACITORS — 30 TOTAL

Reference	Value	Description	Footprint	Qty	Notes
C7 / C18 / C29 / C40 / C51	2.2 nF	SK feedback C_FB (CH0-4)	0402	5	C0G/NP0 mandatory
C9 / C20 / C31 / C42 / C53	4.7 nF	SK shunt C_SH (CH0-4)	0402	5	C0G/NP0 mandatory
C_BP1 — C_BP16	100 nF	TL072 supply bypass (×2 per IC)	0402	16	X7R acceptable — place within 1 mm of VCC/VEE pins
C_BLK1 / C_BLK2	10 μF	Bulk bypass ±9 V rails	0805	2	X7R or tantalum
C_BLK3 / C_BLK4	10 μF	Bulk bypass ±2.5 V ADC rails	0805	2	X7R
Capacitor subtotal				30	

DIODES — 20 TOTAL

Reference	Value	Description	Footprint	Qty	Notes
D2 / D13 / D24 / D35 / D46	BAT54	Input clamp to +9 V (CH0-4)	SOD-323	5	Nexperia BAT54W or PMEG2005
D3 / D14 / D25 / D36 / D47	BAT54	Input clamp to −9 V (CH0-4)	SOD-323	5	
D10 / D21 / D32 / D43 / D54	BAT54	Output clamp to +2.5 V (CH0-4)	SOD-323	5	Chua ±8 V → ±2.0 V within ADS1262 range
D11 / D22 / D33 / D44 / D55	BAT54	Output clamp to −2.5 V (CH0-4)	SOD-323	5	
Diode subtotal				20	

INTEGRATED CIRCUITS — 8 TOTAL

Reference	Value	Description	Footprint	Qty	Notes
U1 — U8	TL072	Dual JFET op-amp (15 signal units + 1 spare)	SOIC-8	8	TL082 or OPA2134 pin-compatible. U8B spare.
IC subtotal				8	
GRAND TOTAL				83	

IC UNIT ASSIGNMENT

Unit	IC	Ch	Function
U1A	U1	CH0	Input buffer
U1B	U1	CH0	Divider buffer
U2A	U2	CH0	Sallen-Key active
U2B	U2	CH1	Input buffer
U3A	U3	CH1	Divider buffer
U3B	U3	CH1	Sallen-Key active
U4A	U4	CH2	Input buffer
U4B	U4	CH2	Divider buffer
U5A	U5	CH2	Sallen-Key active
U5B	U5	CH3	Input buffer
U6A	U6	CH3	Divider buffer
U6B	U6	CH3	Sallen-Key active
U7A	U7	CH4	Input buffer
U7B	U7	CH4	Divider buffer
U8A	U8	CH4	Sallen-Key active
U8B	U8	—	SPARE

FILTER DESIGN (PER CHANNEL)

Parameter	Value
Topology	Unity-gain Sallen-Key 2nd-order Butterworth LPF
R_SK1 = R_SK2	10 kΩ
C_FB (node-A → out)	2.2 nF C0G
C_SH (+in → GND)	4.7 nF C0G
Q factor	0.731 (Butterworth ≈ 0.707)
Corner freq f _n	4.95 kHz
Signal attenuation	÷4 (−12 dB) via 30k/10k divider
Chua ±8 V → ADC	±2.0 V (within ADS1262 ±2.5 V)

$f_n = 1 / (2\pi \cdot R \cdot \sqrt{C_{FB} \cdot C_{SH}})$
 $Q = \sqrt{C_{SH} / C_{FB}} / 2 = \sqrt{4.7/2.2} / 2 = 0.731$